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ASSIGNMENT NO-4

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Problems on Taylor Series method

Q1] Solve $\frac{dy}{dx} = x^2 - y$ with $x_0 = 0$ and $y_0 = 1$

We have $y' = x^2 - 1$ $\therefore y_0' = 0 - 1 = -1$

$y'' = 2x - y'$ $\therefore y_0'' = 2(0) - (1) = 1$

$y''' = 2 - y''$ $\therefore y_0''' = 1$

$y^{IV} = -y'''$ $\therefore y_0^{IV} = -1$

By taylor series method we have

$$y(x) = y_0 + xy_0' + \frac{x^2}{2!}y_0'' + \frac{x^3}{3!}y_0''' + \cdots$$

$$y(x) = 1 + x(-1) + \frac{x^2}{2}(1) + \frac{x^3}{6}(1) + \frac{x^4}{24}(-1) + \cdots$$

$$y(x) = 1 - x + \frac{x^2}{2} + \frac{x^3}{6} - \left(\frac{x^4}{24}\right) + \cdots$$

At $x = 0.1$ we have,

$$y(0.1) = 1 - 0.1 + \frac{(0.1)^2}{2} + \frac{(0.1)^3}{6} - \frac{(0.1)^4}{24} = 0.905163$$

Q2] Solve $\frac{dy}{dx} = 2y + 3e^x$ with $y_0 = 0$

Soln We have, $y' = 2y + 3e^x$ $\therefore y_0' = 0 + 3 = 3$

$y'' = 2y' + 3e^x$ $\therefore y_0'' = 2(3) + 3 = 9$

$y''' = 2y'' + 3e^x$ $\therefore y_0''' = 2(9) + 3 = 21$

$y^{IV} = 2y''' + 3e^x$ $\therefore y_0^{IV} = 2(21) + 3 = 45$

By Taylor Series method we have

P.T.O. →

$$y(x) = y_0 + xy_0' + \frac{x^2}{2!}y_0'' + \frac{x^3}{3!}y_0''' + \cdots$$

$$y(x) = 0 + 3x + \frac{x^2}{2}(9) + \frac{x^3}{6}(21) + \frac{x^4}{24}(\text{calculated}) + \cdots$$

$$y(x) = 0 + 3x + \frac{9x^2}{2} + \frac{7x^3}{2} + \frac{15x^4}{8} + \cdots$$

At $x = 0.1$,

$$y(0.1) = 0 + 3(0.1) + \frac{9(0.1)^2}{2} + 7\left(\frac{0.1}{2}\right)^3 + 18\left(\frac{0.1}{8}\right)^4 = 0.3486$$

At $x = 0.2$,

$$y(0.2) = 0 + 3(0.2) + \frac{9(0.2)^2}{2} + 7\left(\frac{0.2}{2}\right)^3 + 18\left(\frac{0.2}{8}\right)^4 = 0.811$$

Q3]

Soln → let $\frac{dy}{dx} = 1 + y^2$ with $x_0 = 0$ & $y_0 = 0$

we have, $y' = 1 + y^2$ $\quad \therefore y_0' = 1 + 0 = 1$

$y'' = 2y \cdot y'$ $\quad \therefore y_0'' = 0$

$y''' = 2(y \cdot y'' + (y')^2)$ $\quad \therefore y_0''' = 2$

$y^{iv} = 2(yy''' + 2y'y'' + y'y'')$ $\quad \therefore y^{iv}_0 = 16$

By Taylor Series (we get)

$$y(x) = y_0 + x y_0' + \frac{x^2}{2!}y_0'' + \frac{x^3}{3!}y_0''' + \cdots$$

~~$y(x) = 0 + x(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(2) + \cdots$~~

$$y(x) = 0 + x + \frac{x^2}{2}(0) + \frac{2x^3}{3!} + \frac{x^4}{4!}(0) + \frac{x^5}{5!}16 + \cdots$$

$$y(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$$

At $x = 0.2$ we have

$$y(0.2) = 0.2 + \frac{(0.2)^3}{3} + \frac{2(0.2)^5}{15} = 0.202709$$

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Q4) ~~find~~

Soln let $\frac{dy}{dx} = 1-2xy$ with $x_0=0$ and $y_0=0$

we have
$y = (1-2xy)$ $\therefore y_0' = 1-0=1$
$y'' = -2y - 2xy'$ $\therefore y_0'' = 0$
$y''' = -4y' - 2xy''$ $\therefore y_0''' = -4$
$y^{IV} = -6y'' - 2xy'''$ $\therefore y_0^{IV} = 0$
$y^{V} = -8y''' - 2xy^{IV}$ $\therefore y_0^{V} = 32$
$y^{VI} = -10y^{IV} - 2xy^{V}$ $\therefore y_0^{VI} = 0$
$y^{VII} = -12y^{V} - 2xy^{VI}$ $\therefore y_0^{VII} = -384$

By taylor series we get

$$y(x) = 0 + x(1) + \frac{x^2}{2}(0) + \frac{x^3}{6}(-4) + \frac{x^4}{24}(0) + \frac{x^5}{120}(32) + \frac{x^6}{720}(0) + \frac{x^7}{5040}(-384)$$

$$y(x) = x - \frac{2x^3}{3} + \frac{2x^5}{7.5} - 0.762x^2$$

At $x=0.2$

$$y(0.2) = 0.2 - \frac{2(0.2)^3}{3} + \frac{2(0.2)^5}{7.5} - 0.762(0.2)^2$$

$y(0.2) = 0.194752$

At $x=0.4$

$$y(0.4) = 0.4 - \frac{2(0.4)^3}{3} + \frac{2\cdot(0.4)^5}{7.5} - 0.762(0.4)^2$$

$= 0.3598$

Q.5]

Soln let $\frac{dy}{dx} = x+y^2$ and $x_0 = 0$ and $y_0 = 1$

we have $y' = x+y^2$ $\quad \therefore y_0' = 0+1 = 1$

$y'' = 1+2y'$ $\quad \therefore y_0'' = 1+2=3$

$y''' = 2(y')^2 + 2y''$ $\quad y_0''' = 2+6 = 8$

$y^{iv} = 6y'y'' + 2y'''$ $\quad y_0^{iv} = 18+16 = 34$

By Taylor series method we get

$$y(x) = y_0 + xy_0' + \frac{x^2}{2!}y_0'' + \frac{x^3}{3!}y_0''' + \cdots$$

$$y(x) = 1 + x(1) + \frac{x^2}{2}(3) + \frac{x^3}{6}(8) + \frac{x^4}{24}(34) + \cdots$$

$$y(x) = 1 + x + \frac{3x^2}{2} + \frac{4x^3}{3} + \frac{17x^4}{12} + \cdots$$

At $x = 0.2$ we get

$$y(0.2) = 1 + 0.2 + \frac{3(0.2)^2}{2} + 4\frac{(0.2)^3}{3} + \frac{17(0.2)^4}{24}$$

$$y(0.2) = 1.2217$$

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Problems on euler's method

Q.1] given:

Soln $x_0 = 1$

$\frac{dy}{dx} = x+y$ $\quad f(x,y) = x+y$

$y(x_0) = y_0$

$y(0) = 1$

$h = 0.2$

$x_0 = 0$

$y_0 = 1$

$y_1 = y_0 + h\, f(x_0, y_0)$

$y_1 = 1 + 0.2(0+1)$

$y_1 = 1.2$

$y_2 = y_1 + h\,(f(x_1, y_1)$

$= 1.2 + 0.2(0.2 + 1.2)$

$= 1.2 + 0.2(1.4)$

$= 1.48$

$y_3 = y_2 + h\, f(x_2, y_2)$

$= 1.48 + 0.2(0.4 + 1.48)$

$= 1.856$

$y_4 = y_3 + h\, f(x_3, y_3)$

$y_4 = 1.856 + 0.2(0.6 + 1.856)$

$y_4 = 2.3472$

$y_5 = y_4 + h\, f(x_4, y_4)$

$y_5 = 2.3472 + 0.2(1 + 2.3472)$

$y_5 = 2.97664$

Q2] Given:

Soln → $x = 0.3$

$f(x,y) = xy$

$y(x_0) = y_0 = 1$

$y_0 = 1$

$x_0 = 0$

$h = 0.1$

$y_1 = y_0 + hf(x_0, y_0)$

$= 1 + 0.1 f(x_0, y_0)$

$= 1 + 0.1(0.1 \cdot 1)$

$= 1 + 0$

$x_1 = x_0 + h = 0 + 0.1 = 0.1$

$x_2 = x_1 + h = 0.1 + 0.1 = 0.2$

$x_3 = 0.2 + 0.1 = 0.3$

$y_2 = y_1 + hf(x_1, y_1)$

$= 1 + 0.1(0.1 \times 1)$

$y_3 = y_2 + hf(x_2, y_2)$

$= 1.01 + 0.1(0.2 \times 1.01) = 1.0302$

$y(0.3) = 1.0302$

Q.3]

Soln → let $\frac{dy}{dx} = \frac{y-x}{\sqrt{xy}}$ with $y_0 = 2$, $x_0 = 1$, $h = 0.1$ and $f(x,y) = \frac{y-x}{\sqrt{xy}}$

By eulers method we have

$y_{n+1} = y_n + hf(x_n, y_n)$ $n = 0, 1, 2, 3, 4$

we have,

$y_1 = y_0 + hf(x_0, y_0) = 2 + 0.1\left(\frac{2-1}{\sqrt{2}}\right) = 2.07071$

$y_2 = y_1 + hf(x_1, y_1) = 2.07071 + 0.1\left(\frac{2.07071 - 1.1}{\sqrt{1.1 \times 2.07071}}\right) = 2.135028$

$y_3 = y_2 + hf(x_2, y_2) = 2.135028 + 0.1\left(\frac{2.19508 - 1.2}{\sqrt{1.2 \times 2.135028}}\right) = 2.19344$

$y_4 = y_3 + hf(x_3, y_3) = 2.19344 + 0.1\left(\frac{2.193444 - 1.3}{\sqrt{1.3 \times 2.19344}}\right) = 2.246353$

$y_5 = y_4 + hf(x_4, y_4) = 2.246353 + 0.1\left(\frac{2.246353 - 1.4}{\sqrt{1.4 \times 2.246}}\right) = 2.294628$

$y(1.5) = 2.294628$

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Q4]

Ans let $\frac{dy}{dx} = 1+xy$ with $x_0=0$ and $y_0=0.2$, $h=0.1$, $f(x,y) = 1+xy$

By eulers method we have

$y_{n+1} = y_n + hf(x_0,y_1)$. $n=0,1,2,3$

y_{12} $y_0 + hf(x_0,y_0)$

$= 0.2+0.1(1+0.2(0)) = 0.3$

y_{22} $y_1 + h f(x_1,y_1)$

$= 0.3 + 0.1(1+(0.1\times0.3)) = 0.403$

$y_3 = y_2 + hf(x_2,y_2)$

$= 0.403 + 0.1(1+(0.2\times0.403)$

$= 0.51106$

$y_{42} = 0.51106 + (0.1)(1+0.3\times0.51106) = 0.62639$

$\therefore y(0.4) = 0.62639$

Q.5]

Soln let $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $x_0=0$, $y_0=1$, $h=0.02$ $f(x,y) = \frac{y-x}{y+x}$

By eulers method we have

$y_1 = y_0 + h(f(x_0,y_0)$

$= 1+0.02\left(1-\frac{0}{1}+0\right) = 1.02$

$y_2 = y_1 + hf(x_1,y_1) = 1.02+0.02\left(\frac{1.02-0.02}{1.02+0.02}\right) = 1.0392$

$y_{32} = y_2 + hf(x_2,y_2) = 1.0392 + 0.02\left(\frac{1.0392-0.04}{1.0392+0.04}\right) = 1.0577$

$y_{42} = y_3 + hf(x_3,y_3) = 1.0577 + 0.02\left(\frac{1.0577-0.06}{1.0577+0.06}\right) = 1.0756$

$y_{52} = y_{42} + hf(x_4,y_4) = 1.0756 + 0.02\left(\frac{1.0756-0.06}{1.0756+0.06}\right) = 1.0928$

$y(0.1) = 1.0928$

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Problem on modified euler method

Q1)

Soln Let $\frac{dy}{dx} = x^2 + y$ with $x_0 = 0$, $y_0 = 1$, $h = 0.5$, $f(x,y) = x^2+y$

By eolers ~~modified~~ method we have,

$y_1 = y_0 + h f(x_0, y_0) = .$

$y_1 = 1 + 0.5(0+1)$

$= 1.05$

$y_1^{(0)} = y_0 + \frac{h}{2}[f(x_0,y_0) + f(x_1,y_1)] = 1 + \frac{0.05}{2}[0+1 + (0.05)^2 + y_1]$

$y_1^{(1)} = 1 + (0.025)[1.0025 + y_1] = 1.051312$

$y_1^{(2)} = 1.0513445$

Now, $x_1 = 0.05$, $y_1 = 1.0513445$, $x_2 = 0.1$

$y_2^{(1)} = y_1 + h(x_1, y_1) = 1.0513445 + 05\,(0.05)^2 + 1.0513445)$
$= 1.0403018]$

~~$y_2^{(2)} = y_2 + \frac{h}{2}$...~~

$y_2^{(2)} = 1.0513445 + \frac{0.05}{2}[1.0513445\,(0.05)^2 + (1.0403018)]$

$= 1.1043946$

we have

$y_2^{(3)} = 1.0513446 + \frac{0.05}{2}[(0.05)^2 + (1.0513446)] + [(0.1)^2 + 1.1043946]$

$y(0.1) = 1.104404$

Q.2]

Soln → let $\frac{dy}{dx} = x+3y$

$x_0 = 0$

$y_0 = 1$

$h = 0.1$

$f(x,y) = x+3y$

By eolers method we get,

$y_1 = y_0 + h(f(x_0, y_0))$

$= 1 + 0.1(0 + 3(1))$

$= 1.3$

we get,

$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0,y_0) + f(x_1 y_1)]$

$= 1 + 0.\frac{1}{2}[3 + [0.1 + 3(1.3)]$

$= 1 + 0.\frac{1}{2}[7]$

$y(0.1) = 1.35$

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Q.3] 8m

Soln Given →

let $\frac{dy}{dx} = \log(x+y)$, $x_0 = 1$, $y_0 = 2$, $h = 0.2$

$$f(x,y) = \log(x+y)$$

By eular method we get

$y_1 = y_0 + h f(x_0, y_0)$

$y_1 = 2 + 0.2 (\log(3)$

$= 2.219772$

we get by our modified method,

$$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0 + f(x_1, y_1)]$$

$$= 2 + \frac{0.2}{2}[\log(3) + \log(1.2 + y_1)]$$

$= 2.233817$

$y_1^{(2)} = 2.233199$

$y_1^{(3)} = 2.23210$

$y_1^{(4)} = 2.33210$

$y_{22} = y_2 + h f(x_1, y_2) = 2.32210 + 0.2 \log(0.2 + 2.23210) = 2.479909$

$x_1 = 1.2$, $y_2 = 2.233210$

$y_2^{(1)} = y_1 + \frac{h}{2}[f(x_1, y_1) + f(x_2, y_2)]$

$= 2.232210 + 0.1(\log(3.433210) + \log 1.4 + 2.232117)$

$y_2' = 2.492140$

$y_2^{(2)} = 2.492458$

$y_2^{(3)} = 2.492463$

$y(1.4) = 2.492463$

Q4]

Soln → $f(x,y) = x - y^2$

$x_0 = 0$

$y_0 = 1$

$h = 0.2$

By eulers modified method

$y_1 = y_0 + h f(x_0, y_0) = 1 + 0.2(0 - 1) = 0.8$

we have

$x_0 = 0 \quad x_1 = 0.2 \quad y_0 = 1 \quad y_1 = 0.8$

~~$y_1 = 1 + 0.2(0 - 1) =$~~

$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1)]$

$= 1 + \frac{0.2}{2}[-1 + 0.2 - (y)^2]$

$= 0.856$

$y_1^{(2)} = 0.8462$

$y_1^{(3)} = 0.8480$

$y_1^{(4)} = 0.8480$

$x_1 = 0.2 \; , \; y = 0.8480$

Now :-

$y_2 = y_1 + h[f(x_1, y_1)$

$= 0.848 + 0.2(0.2 - (0.848)^2)$

$= 0.24419$

$y_2^{(1)} = y_1 + \frac{h}{2} f(x_1, y_1) + f(x_1, y_1)]$

$= 0.848 + 0.1[0.2 - (0.848)^2 + 0.4 - y_2)^2] = 0.7802$

$y_2^{(2)} = 0.77525$

$y_2^{(3)} = 0.7759$

$y_2^{(4)} = 0.7758$

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Q.5]

Soln → let $\frac{dy}{dx} = 2 + \sqrt{xy}$ with $x_0 = 1.2$

$y_0 = 1.6403$

$h = 0.2$

$f(x,y) = 2 + \sqrt{xy}$

By euler's modified method

$y_1 = y_0 + h f(x_0, y_0) = 1.6403 + 0.2$

$y_1 = 1.6403 + 0.2 (2 + \sqrt{1.2(1.6403)})$

$= 2.32095$

we have $x_0 = 1.2$, $y_0 = 1.6403$ $x_1 = 1.4$

$y_1^{(1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1)]$

$= 1.6403 + \frac{0.2}{2} [(2 + \sqrt{1.2 \times 1.6403}) + (2 + \sqrt{1.4 \times y_1})]$

$= 2.36086$

$y_1^{(2)} = 2.3624$

$y_2^{(1)} = 2.3624$

Now

$x_1 = 1.4$, $x_2 = 1.6$ $y_1 = 2.3624$

$y_2 = y_1 + h f(x_1, y_1)$

$= 2.3624 + 0.2 (2 + \sqrt{1.4 \times 2.3624}) = 3.1261$

$y_2^{(1)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2)]$

$= 2.3624 + 0.1 [2 + \sqrt{1.4 \times 2.3624} + 2\sqrt{1.6 \times y_2}]$

$y_2^{(2)} = 2.96219$

$y_0^{(3)} = 2.96191$

$y_2^{(4)} = 2.961955$

$y(1.6) = 2.961955$